

An Open-architecture AI Model for CPT Coding in Breast Surgery

Development, Validation, and Prospective Testing

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Objective: To develop, validate, and prospectively test an open-architecture, transformer-based artificial Intelligence (AI) model to extract procedure codes from free-text breast surgery operative notes.

Background: Operative note coding is time-intensive and error-prone, leading to lost revenue and compliance risks. Although AI offers potential solutions, adoption has been limited due to proprietary, closed-source systems lacking transparency and standardized validation.

Methods: We included all institutional breast surgery operative notes from July 2017 to December 2023. Expert medical coders manually reviewed and validated surgeon-assigned Current Procedural Terminology (CPT) codes, establishing a reference standard. We developed and validated an AI model to predict CPT codes from operative notes using 2 versions of the pretrained GatorTron clinical language model: a compact 345 million-parameter model and a larger 3.9 billion-parameter model, each fine-tuned on our labeled data-set. Performance was evaluated using the area under the precision-recall curve (AUPRC). Prospective testing was conducted on operative notes from May to October 2024.

Results: Our data set included 3259 operative notes with 8036 CPT codes. Surgeon coding discrepancies were present in 12% of cases (overcoding: 8%, undercoding: 10%). The AI model showed strong alignment with the reference standard [compact version AUPRC: 0.976 (0.970, 0.983), large version AUPRC: 0.981 (0.977, 0.986)] on cross-validation, outperforming surgeons (AUPRC: 0.937). Prospective testing on 268 notes confirmed strong real-world performance.

Conclusions: Our open-architecture AI model demonstrated high performance in automating CPT code extraction, offering a scalable

and transparent solution to improve surgical coding efficiency. Future work will assess whether AI can surpass human coders in accuracy and reliability.

Key Words: artificial intelligence, natural language processing, coding, billing, workflow efficiency, open-source, transformers, operative notes, CPT codes, automation

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Accurate surgical coding is essential for appropriate reimbursement, compliance, and research integrity. However, operative note coding is a labor-intensive process performed by trained medical coders who manually review surgeons' dictated reports and assign the appropriate Current Procedural Terminology (CPT) codes. This process is particularly expensive,¹ and is also prone to errors, leading to missed billing opportunities or overbilling, both of which carry significant financial implications. The Centers for Medicare & Medicaid Services estimates that in 2024, the Medicare Fee-for-Service program had \$31.7 billion in improper payments.² Among improper payments in general surgery, 36% were attributed to coding errors.² These inefficiencies increase administrative overhead and delay revenue cycle operations, particularly in high-volume surgical practices.^{3,4} Beyond financial losses, coding errors may compromise the quality of procedure data recorded in research databases.

Artificial intelligence (AI) has shown promise in automating medical coding, particularly for extracting ICD-10 codes from free-text clinical notes, such as discharge summaries⁵ and trauma tertiary survey notes.⁶ Commercial AI-based tools have been developed to assist medical coders in CPT coding by automating billing code extraction from clinical documentation, reducing errors, and improving billing turnaround times.^{7–9} Despite these benefits, the adoption of such tools remains limited due to their proprietary, closed-source nature. These algorithms are often complete “black boxes” in multiple respects: (1) their development and training processes are opaque, (2) their architecture and weights are proprietary, (3) they cannot be audited or adapted to local use, (4) they lack standardized validation, which is crucial for ensuring reliability in clinical settings,¹⁰ and (5) like most complex AI models, their internal logic is difficult to interpret, limiting understanding of how predictions are made. Furthermore, licensing and implementation can be substantial, which may limit accessibility for some institutions. This can pose challenges for smaller or resource-constrained systems and contribute

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The de-identified data underlying this study contain protected health information and are subject to institutional and regulatory requirements. Consequently, these data are not publicly available. However, interested researchers may request access to a limited, de-identified data set by contacting the corresponding author. Requests will be reviewed on a case-by-case basis and may require appropriate data-use agreements and Institutional Review Board approval in accordance with institutional policies.

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to disparities in administrative efficiency. Finally, the integration of AI into clinical workflows may require significant changes in data management and workflow integration, which can be challenging for health care organizations.¹⁰

Given these limitations, there is a clear opportunity to develop transparent, open-source AI solutions that democratize access to advanced operative note coding tools. Unlike proprietary systems, open-source models offer full transparency: their architecture, training data, and learned weights are publicly accessible, giving users complete control over how the model is evaluated, deployed, and updated. Institutions can fine-tune these models to reflect evolving coding guidelines and adapt them to their specific documentation standards.¹¹ Although large AI models remain difficult to interpret due to their size and multi-layered structure, even when open-source, access to their weights allows researchers to probe model behavior and gain partial insight into which components contribute to specific coding decisions.^{12–15} These features facilitate reproducibility, peer review, and external validation, thus promoting trust and accountability in medical coding practices.¹⁶ In turn, open-source models can significantly streamline billing processes, minimize coding errors, and mitigate backlogs, thereby enhancing productivity and accuracy in surgical documentation workflows. Although preliminary research has explored AI-based CPT code extraction from free-text operative notes,¹⁷ a validated,

industry-standard open-source AI tool specifically designed for surgical operative notes is still lacking.

To address this gap, we developed, validated, and prospectively tested a novel, transparent, open-architecture transformer-based AI model for automatically extracting CPT codes from free-text breast surgery operative notes. We hypothesize that such a model is feasible and will achieve exceptionally high performance, comparable to that of expert medical coders. This study focuses on evaluating how closely AI's predictions align with human coders, establishing a foundation for future work aimed at determining whether AI can ultimately surpass coders in accuracy and efficiency. Its open-architecture design ensures transparency, allowing for external auditing and trust in AI decision-making. By optimizing computational efficiency, this model is scalable and suitable for diverse clinical settings. Furthermore, real-world validation, including prospective testing, confirms its clinical applicability and reliability.

METHODS

Our study was exempt from the University of Virginia Institutional Review Board. This study adhered to the Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD) guidelines (Supplemental Table 1, Supplemental Digital Content 1, <http://links.lww.com/SLA/F509>).

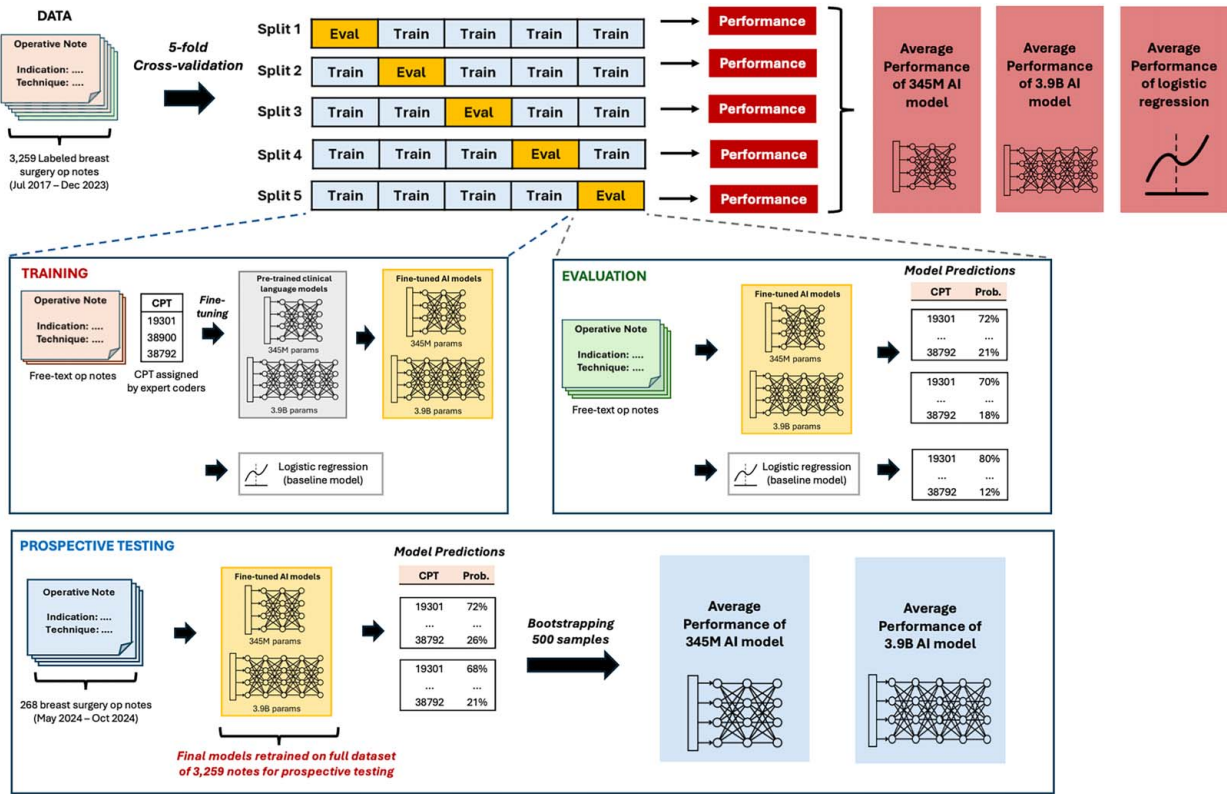


FIGURE 1. Overview of the development and validation of AI models for CPT code prediction from operative notes. This schematic illustrates the end-to-end workflow for developing and evaluating AI models for CPT code prediction from breast surgery operative notes. Model performance was estimated using 5-fold cross-validation for logistic regression, a 345M-parameter fine-tuned clinical language model, and a larger 3.9B-parameter model. Metrics were averaged across folds. Final models were retrained on the full data set and evaluated on prospectively collected operative notes. Performance on the test set was assessed using bootstrap resampling.

Data Source

We queried our institutional database and retrieved all breast surgery operative notes from the University of Virginia Medical Center, covering the period from July 2017 through December 2023. Notes were identified based on procedures performed by surgeons in the Division of Breast Surgery. The corresponding dictated notes were extracted from the institutional electronic health record system using structured SQL queries and linked to the CPT codes assigned by certified professional medical coders using the patient's medical record number, procedure date, and time. Notes lacking surgeon-provided CPT codes were excluded. Each operative note was manually reviewed by expert medical coders, who validated and corrected the CPT codes provided by surgeons, establishing a robust reference standard for model evaluation. Surgeon-provided CPT codes were compared against this validated reference standard to evaluate coding discrepancies, specifically identifying rates of overcoding (extraneous codes assigned that did not correspond to actual performed procedures) and undercoding (performed procedures that were omitted from the provided CPT codes).

Operative Note Processing

All operative notes were fully de-identified before analysis using a regular expression-based script followed by manual spot checks for quality assurance. To maintain consistency and relevance, only the 'procedure' portion of each operative note (which explicitly details the procedural steps and methods) was included. Other sections, such as procedure indication, surgeon-assigned CPT codes, and specimen descriptions, were excluded to avoid data leakage and ensure the AI model focused solely on operative technique descriptions.

Model Development

Figure 1 provides an overview of the model development and validation process. The ground truth consisted of CPT codes that were originally assigned by certified medical coders at the time of clinical care. These codes were not re-reviewed or modified, to reflect real-world coding practices currently implemented in clinical settings. We excluded CPT with fewer than 10 occurrences in our data set as infrequent codes introduce statistical instability due to insufficient training examples and hold limited clinical relevance due to their rarity and minimal financial impact.

Operative notes were tokenized with a maximum sequence length of 512 tokens, which corresponds to the 85th percentile of note lengths in our data set. This balanced computational complexity with adequate representation of note content. A logistic regression model using Term Frequency-Inverse Document Frequency (TF-IDF) features was initially trained as a baseline to assess the predictive capability of a simpler, linear method for CPT code extraction, establishing a reference point against which the added performance benefit of an AI model could be objectively evaluated. We then developed 2 AI models to predict CPT codes from free-text operative notes. Each model starts with GatorTron, an open-source transformer-based clinical language model,¹⁸ and adds a fully connected output layer that assigns probabilities to each CPT code for every operative note. The first version is a compact 345-million-parameter model, optimized for computational efficiency. The second is a larger 3.9-billion-parameter

model built to evaluate whether increased complexity improves predictive performance.

To ensure robust performance estimates, we implemented 5-fold cross-validation for the logistic regression and AI models. The data set was split into 5 folds, with each fold used once for validation and the remaining 4 for training. AI models were trained using binary cross-entropy loss and prespecified hyperparameters (Supplemental Table 2, Supplemental Digital Content 1, <http://links.lww.com/SLA/F509>). TF-IDF features for the logistic regression model were generated only from the training folds. Performance metrics were averaged across the 5 folds, and 95% CIs were calculated using standard error and the *t*-distribution. After cross-validation, the AI models were retrained on the full data set using the same hyperparameters for prospective testing. Development was performed locally using Python with PyTorch and HuggingFace's Transformers library on an NVIDIA A100 GPU within our institution's secure high-performance computing cluster. This approach added an additional layer of data security, despite all notes being de-identified.

Performance Evaluation

Model and surgeon performance were assessed using the area under the receiver operating characteristic curve (AUC), precision (percentage of correctly predicted codes), recall (percentage of actual codes captured), and F1 score (harmonic mean of precision and recall). Since AUC can be misleading in multilabel settings with class imbalance, as it may overstate model performance by emphasizing true negatives, we relied on area under the precision-recall curve (AUPRC) as the primary measure of overall performance. AUPRC better captures model behavior in imbalanced data sets. Given the multilabel nature and varying frequency of CPT codes, AUPRC, F1 score, and AUC were reported using micro-averaging and weighted-averaging methods. Micro-averaging calculates performance across all CPT code predictions combined, meaning that more frequent codes have a greater impact on the final metric, whereas weighted-averaging adjusts for label frequency, ensuring that performance metrics reflect the true distribution of CPT codes in the data set rather than being dominated by the most common ones. Macro-averaging, which gives equal importance to all codes regardless of how rare they are, was not computed, as rare CPT codes are less clinically and financially relevant and can skew the results. Precision and recall were reported as overall metrics across all predictions.

Prospective Testing

To evaluate the generalizability and predictive reliability of the final model beyond retrospective data, we prospectively tested the AI models on all breast surgery operative notes dictated at our institution between May 2024 and October 2024. The model was only trained to predict the 19 CPT codes included in the training data. As a result, it could not predict rare codes outside this label set, and these were not included as target labels during evaluation. We did not penalize the model for failing to predict codes it had not seen during training, consistent with standard practice in multilabel classification. To assess the statistical uncertainty of model performance on the prospective test set, we performed bootstrap resampling with 500 iterations to generate 95% confidence intervals for all performance metrics.

RESULTS

Our final analytic data set included 3259 operative notes with a total of 8036 CPT codes. This data set excluded 52 CPT codes that appeared fewer than 10 times, spanning 49 operative notes (Supplemental Table 3, Supplemental Digital Content 1, <http://links.lww.com/SLA/F509>). Each operative note contained a median of 2 CPT codes (range: 1–8), with a median word count of 300 (range: 90–1570). There were 19 unique CPT codes, and the frequency of each code is presented in Table 1. There were discrepancies in 12% (n = 380) of notes when comparing surgeon-provided CPT codes against the expert-validated reference standard. Overcoding was observed in 8% (n = 255) of notes, whereas undercoding was found in 10% (n = 309) of notes.

Model Performance on Cross-validation Data

Table 2 presents the fine-tuned transformer-based models’ performance against the reference standard in the cross-validation set. Both AI models outperformed the baseline logistic regression model across most evaluation metrics. The compact model (345M parameters) achieved a weighted AUPRC of 0.976 [0.970, 0.983], weighted F1 score of 0.954 [0.940, 0.967], precision of 97.0% [95.8%, 98.3%], and recall of 94.7% [93.8%, 95.7%], whereas the larger model (3.9B parameters) demonstrated a weighted AUPRC of 0.981 [0.977, 0.986], weighted F1 score of 0.961 [0.957, 0.965], precision of 97.1% [96.6%, 97.6%], and recall of 95.6% [94.9%, 96.2%]. In contrast, the logistic regression model exhibited lower performance across most metrics, with a weighted AUPRC of 0.956 [0.948, 0.964], weighted F1 score of 0.884 [0.875, 0.894], a precision of 95.4% [94.7%, 96.2%], and a recall of 86.7% [85.5%, 87.9%].

The AI models achieved higher overall performance compared with surgeons. Although their weighted F1 scores were slightly lower, this metric depends on a fixed prediction threshold (set at 50% in our analysis). In contrast, the weighted AUPRC, which reflects model precision across all thresholds, was significantly higher for both AI models. This indicates that the AI models were more consistently accurate in identifying the correct CPT codes, regardless of the sensitivity level selected.

Performance varied by CPT code, as illustrated in Figure 2. The highest-performing CPT codes, such as simple mastectomy, sentinel lymph node mapping, and sentinel lymph node biopsy achieved near-perfect AUPRC and F1 scores. These CPT codes were also among the most frequently occurring in the data set (Table 1). In contrast, lower-performing CPT codes included intraoperative radiotherapy applicator placement and certain mastectomy procedures, which demonstrated low AUPRC and F1 scores. These codes were among the least frequently recorded in the data set.

Prospective Testing

The final AI models trained on the full data set after cross-validation were prospectively tested on 268 operative notes and demonstrated robust performance with near-perfect predictions closely matching expert medical coders’ CPT assignments. The compact model achieved a weighted AUPRC of 0.987 [0.978, 0.996] and F1 score of 0.973 [0.960, 0.984] whereas the larger model performed similarly, with a AUPRC of 0.991 [0.987, 0.994] and weighted F1 score of 0.973 [0.967, 0.979] (Table 3), with both models outperforming surgeons (weighted AUPRC: 0.937, weighted F1: 0.972).

TABLE 1. Distribution of CPT Codes Included in Model Training and Prospective Testing Data Sets

CPT code	CPT code description	Frequency in Cross-validation set (n = 8036), n (%)	Frequency in prospective testing set (n = 612), n (%)
19301	Partial mastectomy	1484 (18.5)	141 (23.0)
38900	Intraoperative sentinel LN mapping with dye injection	1368 (17.0)	82 (13.4)
38525	Deep axillary LN biopsy/excision	1321 (16.4)	82 (13.4)
38792	Intraoperative sentinel LN mapping with radioactive tracer	850 (10.6)	61 (10.0)
19303	Simple mastectomy	732 (9.1)	56 (9.2)
14001	Adjacent tissue transfer (10–30 cm ²)	623 (7.8)	100 (16.3)
19125	Excision of breast lesion identified by preoperative placement of radiological marker	536 (6.7)	48 (7.8)
14000	Adjacent tissue transfer ≤ 10 cm ²	370 (4.6)	8 (1.4)
36590	Removal of tunneled CVC with port/pump	139 (1.7)	8 (1.4)
19297	Placement of expandable breast catheter with concurrent partial mastectomy	124 (1.5)	1 (0.2)
19307	Modified radical mastectomy	110 (1.4)	6 (1.0)
19120	Excision of breast lesion	106 (1.3)	9 (1.5)
14301	Adjacent tissue transfer (30–60 cm ²)	91 (1.1)	1 (0.2)
19110	Nipple exploration	58 (0.7)	6 (1.0)
19126	Excision of breast lesion identified by preoperative placement of radiologic marker	32 (0.4)	1 (0.2)
19302	Partial mastectomy with axillary lymphadenectomy	30 (0.4)	1 (0.2)
19294	Placement of a radiation therapy applicator for IORT during partial mastectomy	23 (0.3)	0
19296	Placement of radiotherapy afterloading expandable catheter after partial mastectomy	20 (0.3)	0
38745	Complete axillary lymphadenectomy	19 (0.2)	0

CVC indicates central venous catheter; IORT, intraoperative radiation therapy; LN, lymph node.

TABLE 2. Comparative Performance of AI Model, Surgeons, and Simple Logistic Regression in Predicting CPT Codes on Cross-validation Data

	Surgeons' performance	Logistic regression [95% CI]	Transformer-based AI models [95% ci]	
			Compact model (345M params)	Large model (3.9B params)
F1 Score _{Weighted}	0.964	0.884 [0.875, 0.894]	0.954 [0.940, 0.967]	0.961 [0.957, 0.965]
F1 Score _{Micro}	0.959	0.909 [0.901, 0.916]	0.955 [0.948, 0.970]	0.963 [0.959, 0.968]
AUC _{Weighted}	0.975	0.991 [0.988, 0.994]	0.994 [0.990, 0.998]	0.995 [0.992, 0.998]
AUC _{Micro}	0.978	0.993 [0.991, 0.994]	0.996 [0.993, 0.999]	0.996 [0.993, 0.999]
AUPRC _{Weighted}	0.937	0.956 [0.948, 0.964]	0.976 [0.970, 0.983]	0.981 [0.977, 0.986]
AUPRC _{Micro}	0.926	0.968 [0.963, 0.974]	0.984 [0.976, 0.992]	0.986 [0.979, 0.993]
Precision	95.9%	95.4% [94.7%, 96.2%]	97.0% [95.8%, 98.3%]	97.1% [96.6%, 97.6%]
Recall	96.0%	86.7% [85.5%, 87.9%]	94.7% [93.8%, 95.7%]	95.6% [94.9%, 96.2%]

Bold values highlights the model with the highest performance for each row.

DISCUSSION

Surgical coding remains a time-consuming process with minimal tolerance for error, given the financial and compliance implications of incorrect CPT code assignment.^{19,20} Our study demonstrates that an AI-driven approach can enhance coding accuracy while reducing administrative burdens on surgeons and coders. Our coding tool was developed using an open-source large language model pretrained on medical text corpora, based on the rationale that prior medical language knowledge would enhance the AI's ability to recognize clinical details²¹ such as procedure type, anatomic location, and surgical technique, and facilitate fine-tuning to accurately map these concepts to the correct CPT codes. The model achieved excellent performance, with AUPRC and F1 scores exceeding 96%. By prospectively validating our model, we ensure that it performs reliably in real-world clinical settings, addressing a key limitation of many AI-driven health care solutions. This study represents an important step toward integrating AI into daily surgical workflows in a transparent, scalable, and cost-effective manner.

In this study, we specifically assessed how closely the AI model's predictions aligned with expert medical coders, who served as our reference standard. As such, in this design, a perfect match (100% agreement) would mean the AI is effectively indistinguishable from human coders. Our results show near-perfect alignment between the AI-generated CPT codes and those assigned by expert coders in most operative notes. With AI now established as a highly performant tool that closely mirrors coder predictions, future work will focus on a superiority study to determine whether AI can surpass expert coders by systematically analyzing discrepancies between AI and human coding to identify instances where AI may outperform traditional manual coding.

Furthermore, the AI model outperformed surgeons in overall coding performance. Surgeons' coding generally does not match the accuracy of dedicated coding specialists, due to time constraints and less familiarity with the intricacies of coding rules.^{22,23} Our AI model's AUPRC was about 5% to 6% higher than that of surgeons. Although this improvement may seem modest, it has substantial financial implications when scaled nationally. More than 200,000 mastectomies and lumpectomies are performed annually in the United States,²⁴ and this increase in performance could safeguard a significant portion of

revenue that would otherwise be lost. Specifically, our AI models demonstrated higher precision compared with surgeons, which is particularly valuable for compliance, as the AI model rarely suggests extraneous codes, thus reducing the risk of overbilling and the severe penalties associated with false claims. However, a system optimized for very high precision may occasionally miss legitimate codes (lower recall), leading to underbilling. Despite this tradeoff, our model achieved an impressive recall of 96%. Thus, beyond improving coding accuracy, the AI system offers tangible financial benefits and risk mitigation when deployed at scale.

We explored 2 versions of our AI model, a compact model with 345 million parameters and a larger model with 3.9 billion parameters, and found that both outperformed simple logistic regression model in automatic CPT code extraction. However, increasing AI model complexity did not yield significant performance gains. This aligns with prior research showing that simpler NLP models can sometimes outperform more complex deep learning models for specific coding tasks.²⁵ This finding has important practical implications. The smaller model is lightweight, runs efficiently on standard hardware, and processes notes quickly with minimal computational overhead. It can therefore be deployed in any setting, including resource-limited practices that lack dedicated coders and cannot afford expensive proprietary software. In contrast, the larger model is resource-intensive and requires specialized high-performance computing infrastructure, which is typically available only at large institutions or through purchasing cloud computing time. This represents costs that may be prohibitive for many practices. It also incurs longer processing times per note, and in busy clinical settings, even a few seconds of added latency per case can significantly impede workflow.

The development of an AI coding tool based on an open-source transformer model with local deployment capabilities has important implications for surgical departments and health systems. First, it eliminates the high costs associated with proprietary coding solutions, which often require expensive licenses or subscriptions. In contrast, by open-sourcing our code and leveraging an open-architecture model, we enable institutions and practices to deploy advanced coding assistance locally without additional software fees or costly cloud hosting, thereby making advanced coding assistance accessible even to resource-limited clinics. In addition, open-source ensures greater

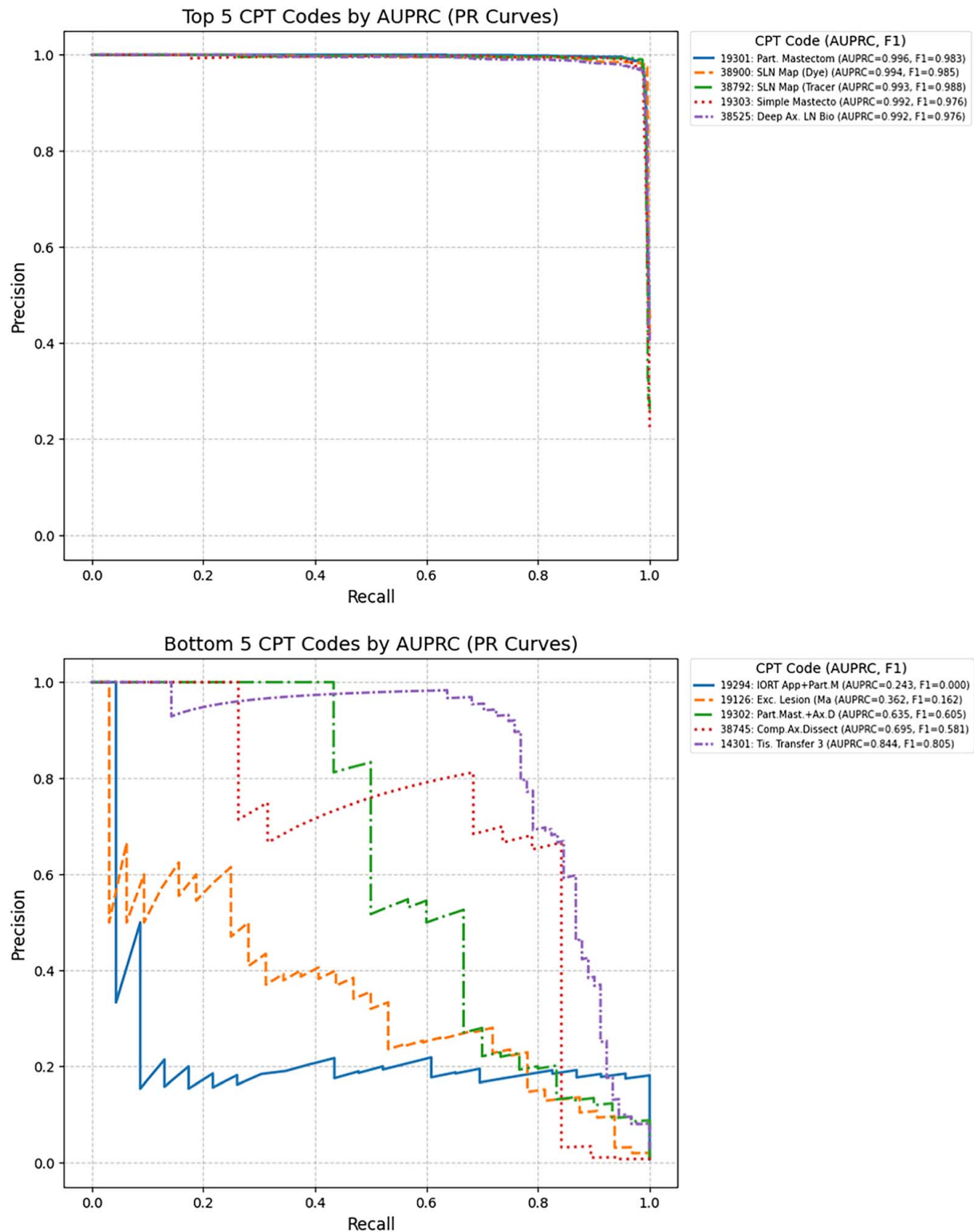


FIGURE 2. Precision-recall curves for the top 5 and bottom 5 CPT codes using the lightweight AI model on the cross-validation set.

transparency compared with proprietary black-box algorithms. Their architecture, training data, and learned parameters can be fully accessed, inspected, and audited, which is crucial in health care where trust and verifiability

are paramount.¹⁶ Although transparency does not make such models inherently interpretable, it does allow researchers to conduct controlled experiments to better understand model behavior, supporting limited forms of

TABLE 3. Performance of AI Model in Predicting CPT Codes on Prospective Data.

Surgeons' performance		Transformer-based AI model [95% ci]	
		Compact model (345M params)	Large Model (3.9B Params)
F1 Score _{Weighted}	0.972	0.973 [0.960, 0.984]	0.973 [0.967, 0.979]
F1 Score _{Micro}	0.972	0.977 [0.966, 0.984]	0.974 [0.968, 0.980]
AUC _{Weighted}	0.982	0.996 [0.992, 0.999]	0.998 [0.997, 0.999]
AUC _{Micro}	0.983	0.997 [0.994, 0.999]	0.999 [0.998, 0.999]
AUPRC _{Weighted}	0.937	0.987 [0.978, 0.996]	0.991 [0.987, 0.994]
AUPRC _{Micro}	0.926	0.990 [0.980, 0.997]	0.994 [0.991, 0.997]
Precision	97.0%	98.8% [97.8%, 99.7%]	98.0% [97.2%, 98.8%]
Recall	96.1%	96.5% [95.0%, 99.7%]	96.9% [95.9%, 97.8%]

Bold values highlights the model with the highest performance for each row.

explainability.^{15–15} Second, the AI's prediction threshold can be adjusted to balance precision and recall, allowing institutions to tailor its performance to their risk tolerance and priorities. A risk-averse practice concerned about overbilling can set a high threshold such that only highly confident CPT code predictions are generated, with the tradeoff that the model might be miss less-obvious codes. Conversely, if maximizing revenue capture is the priority, the threshold can be lowered to favor recall, accepting a minor increase in false positives that can be reviewed manually. This tunability enables coding professionals to use AI as a precision instrument, adjusting its scope as needed, rather than a rigid tool. It also highlights an important principle for AI in health care: flexibility and human oversight are crucial.²⁶ The AI augments, rather than replaces, human coders, allowing them to transition into a supervisory role where they set parameters, validate outputs, and focus on complex cases. This is particularly valuable for rare procedures where training data are limited and experienced human judgment is needed.

Although this study focused on model development and validation, we also created a simple point-and-click interface to demonstrate potential integration into clinical workflows. Given the minimal performance difference between the compact and large models, we selected the 345M model for deployment due to its greater efficiency, ease of use, and balance of speed, portability, and accuracy. The interface allows users to upload operative notes, view predicted CPT codes with confidence scores, and adjust the probability threshold to control model sensitivity. Although not formally evaluated, the tool serves as a proof of concept for human-in-the-loop workflows, enabling coders to streamline verification while retaining oversight. Future work will evaluate coder interaction with the tool, including usability, workload, and effectiveness in real-world settings.

This study has some limitations that we acknowledge. First, the AI model was trained on data from a single hospital and therefore may not generalize well to other institutions. Future work should incorporate external data sets to improve robustness and ensure broader clinical applicability. Second, our reference standard for “ground truth” CPT codes was based on expert human coders' work, which is not infallible. If the AI predicted a code that the original coder missed, our evaluation would count the AI as “wrong” when in fact it might have identified a legitimate billing opportunity overlooked in the manual process. Although we did not have a mechanism to adjudicate such

discrepancies in this study, ongoing work is being undertaken to systematically review coding mismatches to determine whether the AI model can outperform human coders in specific contexts. Third, although our model showed strong overall performance, it struggled with rare procedures that were underrepresented in the training data. In such cases, the model might default to suggesting a nearest-known code or fail to output a code entirely. This limitation is inherent to data-driven AI systems, which are constrained by the scope of their training example. In addition, we excluded extremely rare CPT codes (<10 occurrences) from model training and evaluation. Although this could slightly limit generalizability, these codes represented <1% of all CPTs and are unlikely to meaningfully bias performance estimates. Fourth, our study was limited to breast surgery operative notes, which exhibit relatively high uniformity in both documentation and coding. Surgical specialties with more complex coding requirements, such as vascular surgery, require further evaluation to assess the AI model's performance in coding scenarios with greater procedural variability. Finally, our model did not integrate CPT modifiers, as this pilot study focused on baseline coding accuracy. Ongoing work aims to incorporate modifiers and expand the scope of the model beyond breast surgery to a broader range of procedures.

Beyond these efforts, developing a truly general-purpose surgical coding model will require the implementation of additional techniques to address persistent challenges, particularly performance limitations on low-frequency procedures. Such strategies include data augmentation, such as paraphrasing operative descriptions to diversify rare examples; synthetic operative note generation, where large language models can be prompted to produce realistic notes for infrequent procedures based on CPT descriptions; and few-shot prompting, in which the model is guided during inference by a small number of annotated examples to improve prediction accuracy without the need for extensive retraining.

CONCLUSIONS

By demonstrating high performance, real-world feasibility, and broad accessibility, this study provides a strong foundation for the adoption of open-source AI in surgical coding. The ability to automate CPT code extraction with minimal computational overhead has the potential to reduce administrative workload, improve billing accuracy, and enhance efficiency across health care systems. As AI

continues to evolve, transparent and scalable models like this one will be crucial in ensuring equitable access to advanced medical coding solutions, ultimately driving improvements in both clinical and financial outcomes.

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DISCUSSANT

Discussion of Paper Number 17

Development, Validation, and Prospective Testing of a Novel Open Architecture AI Model for Automated Surgical Coding

Dr. Genevieve B. Melton-Meaux (Minneapolis, MN)

Thank you. I have no relevant disclosures. It's my pleasure to discuss this work from Drs El Moheb, Tsung and team. I appreciate the authors providing the review of their manuscript in advance.

The authors describe the training, validation, and prospective evaluation of an open-source language model to determine CPT codes in breast cancer surgery. Their language model leverages a previously developed health care language model called Gatortron, and then it was fine-tuned with 6 years of operative reports using both a more compact and then a more complex approach. And, overall, they showed good performance with the compact approach and saw no significant improvement using the more complex approach.

It's a well-conceived study, and it demonstrates the potential of these approaches to bring substantial efficiencies to administrative tasks like procedure coding and registries. Today, for instance, the American College of Surgeons is in the evaluating and moving towards the use of artificial intelligence and machine learning to power insights and data collect for its registries. This study is a clear demonstration

of how natural language processing with the use of language models is revolutionizing revenue cycle data abstraction and other health care processes.

I have several questions. First, there were instances, as you indicated, of under and overcoding observed by the language models. You used expert medical coders for ground truth. With this, can you comment on the adjudication that was made around this and how discrepancies were looked at since human coders can have errors?

Second, one element in the paper, as far as methodologies, was that the authors eliminated procedures that had 10 or less examples, and these were excluded from both the training and also possibly from the validation data set, including possibly the prospective evaluation data sets. So, with these procedures excluded, is the reported performance overstated? And along those lines, how many of the procedures were excluded with these rarer examples?

And then finally, when you look towards the future, how do the authors envision expanding and scaling these methods beyond breast surgery, and how might the issues that you encounter with scarcity of data be overcome?

Again, I want to commend you and the team for this excellent work, and I appreciate the opportunity to comment.

Response from Mohamad El Moheb

Thank you very much, Dr Melton-Meaux, for discussing our paper and for raising such important questions.

Regarding your first question about how we adjudicated discrepancies between AI and expert coders, this is the next step in our study, and we are currently in the process of doing that. In this study, our goal was to assess how closely AI performance could align with expert coders, who were used as the reference standard.

We assumed the coder-assigned CPT codes were correct. But coders can make mistakes. Determining whether AI or coders are superior would require establishing a different reference point, which is challenging because the codes assigned by expert coders are the ones submitted for billing and insurance, effectively making them the current gold standard.

To address this, third-party coders who were not involved in the original coding will review the discrepancies between AI and human coders. They will receive a blinded set of CPT codes—one set generated by AI and the other by the original coders—and will be asked to determine which set is more appropriate and provide their rationale. We hope this will help us better understand whether AI is truly matching or even surpassing coder-level performance.

As for your second question about rare codes, I should have mentioned that excluded codes comprised a very small proportion of our data set. Out of ~8000 total CPT assignments, <1%, or about 50 codes, across 30 operative notes were removed. This was necessary to reduce statistical noise. Many of these rare codes had only a single instance, which means that we can either train the model on them or validate on them, but not both.

For prospective testing, we excluded codes that the model was not “exposed to” during the training phase since these codes fall outside the model’s prediction space. To address this limitation in future studies, and this relates to your final question, we intend to use data augmentation techniques to increase representation of rare codes. This involves generating synthetic operative notes so that the model learns from a broader variety of writing styles and improves its ability to generalize. It should also increase the

robustness of predictions, particularly for underrepresented procedures.

Finally, regarding expanding the model to other surgical specialties, this is exactly what we are working on now. Instead of building separate models for each specialty one at a time, which would be very time-consuming, we will scale the model across multiple specialties at once and then assess performance to see where the model performs well and where it struggles. I suspect that for specialties that have uncommon procedures, we will need to rely on augmentation techniques to make the model reach more robust predictions. Thank you.

Dr. Hasan Alam (Chicago, IL)

I love this work. I think this is the future. Pretty soon, we will have AI doing a lot of these repetitive tasks efficiently and quickly. I can also see that the payers would deploy their own AI programs to review your data and potentially challenge your coding decisions.

You can almost imagine it turning into a war of the AI programs, with your programs billing and their AI programs reviewing and denying your bills. So, if you think about it, how is this going to be resolved? Who would decide what is the truth when looking at the recommendation generated by AI programs designed for different goals?

Similarly, if you think about it, apart from the financial consequences, there are medical legal consequences. Institutions are held accountable for accuracy of the data. If you are caught overbilling, there are penalties and financial consequences.

In your opinion, who would be responsible for the accuracy of the AI-generated billing? Is it going to be the surgeon, the person who’s designed the AI program, your IT department, the billing office? Theoretically, there may not even be a person left in the billing office if the AI program is handling all the billing.

Overall, I think it is excellent work, and I have no doubt that this is the future.

Response from Mohamad El Moheb

Thank you very much, Dr Alam. Your first question is an excellent point. I think we will reach an equilibrium when practices develop increasingly robust models that generate more accurate codes and insurance companies do the same. Eventually, this kind of adversarial system will push the performance of both models, and they will reach an equilibrium.

This is a good example of how game theory might apply. When both models reach this equilibrium, the codes will ideally match on both the practice side and the insurance side, which will hopefully reduce the need for auditing.

Your second question regarding the medico-legal consequences is also an excellent point. Even as we develop highly accurate AI tools, I do not see these methods replacing coders in the near future, because there needs to be a shift in both culture and mindset. Right now, almost all AI applications require a human in the loop to verify the accuracy of generated codes and to ensure accountability.

That said, as we continue to publish studies like this and demonstrate the robustness of AI methodologies, and as surgeons and practices begin to gain more trust in these tools, we hope to gradually automate many of these repetitive tasks.

Dr. William Richards (Mobile, AL)

Great talk. I have no relevant disclosures.

I dislike your premise. Your premise is that the coders are better than the surgeons, okay? So, number one, I think the surgeons are much better than the coders. I argue with my coders all the time.

Furthermore, I know what procedure I did. I did a sentinel lymph node biopsy. I did a partial mastectomy as well. And I did it on both breasts because there was bilateral cancer.

What I need to know is how do I write a note that documents all those and puts in all these arcane modifier codes, so I maximize my work RVUs and document the work that I performed.

So, in other words, let's turn it around. We don't know all these arcane little things that you need to say in your operative note. Have AI help us put in the note of what we need to say to document it, okay?

But AI is the future. And get away with the coders because I argue with them all the time, and they're frequently wrong.

Response from Mohamad El Moheb

Thank you very much for your questions, and I apologize if I gave the impression that coders are better than surgeons. That was certainly not my intention.

You raised a very important point, which is how we can make sure that the operative notes we are writing include sufficient documentation to justify certain CPT codes. This is especially relevant for the 22 modifier.

When we were first brainstorming ideas for this project, this was one of the first things we discussed. One direction we are exploring for the future is integrating a visual, self-attention mechanism that highlights which parts of the operative note the model is focusing on when predicting a code.

This could help surgeons understand whether the note includes enough language to justify the inclusion of a particular CPT code. This is something we are very interested in, and we hope future iterations of the model will include this feature.

Dr. Michael D'Angelica (New York, NY)

It's a very similar question. I think there's no question the computing power will be able to handle this and there is no doubt that it's the future. It's really present in a lot of industries right now. And it'll only get better, and this will be the easy part of this.

But no matter how good your computing power is, garbage in always equals garbage out. So, this has everything to do with the operative note. Because if you're not documenting what you do in some way in that note, you'll never capture it.

So, my question really is are you considering looking at things like synoptic operative notes and ways that language can be standardized? Because if you think about it, most of what we do is the same old CPTs over and over again. There is no reason the language has to be any different for any of those notes, aside from maybe some clinical details that are relevant to you and the people who read your notes.

But from a billing perspective, are you looking into sort of standardizing language for it? It's a similar question, but I think that's the key.

Response from Mohamad El Moheb

Thank you very much. This is one of the main reasons we chose to focus on breast surgery for this initial study. There were 2 key factors behind that decision.

First, breast surgeons at our institution write their own operative notes. Second, those notes are highly standardized. The surgeons meet regularly with coders to review documentation practices and get feedback on the language needed to justify specific CPT codes.

As a result, their operative notes use consistent templates and dot phrases to ensure high-quality documentation. This is both a strength and a limitation. The strength is that we have access to clean, standardized, high-quality data. But the limitation is that it is possible that the AI model is learning to predict specific CPT codes based on very specific phrasing. That means it could struggle to generalize if the same procedure is documented in a different way.

This ties back to my earlier comment about using data augmentation. We need to diversify the phrasing, tone, and structure of the operative notes so that the model can recognize and correctly assign CPT codes even when the wording changes. And that is not just important for breast surgery, but it will be important to other surgical specialties as well.

Dr. Chandrakanth Are (Omaha, NE)

Thank you. Congratulations on your fine presentation and the mastery on the topic. I have 2 questions.

One, we know we're somewhere between hype and reality with AI. So, through all of that, there are the ethical aspects of AI. For your study, when you used real patients, were there any HIPAA concerns, and how did you address them? Because when you put data, particularly into open source, you don't know who else can have access to that.

Two, even at its best, sometimes AI is not accurate. So, for your study, did you repeat the analysis again? Because you ask AI the same question multiple times, it's quite likely you'll get a different answer each time. Thank you.

Response from Mohamad El Moheb

Thank you. Two very excellent questions. For the first one, we used a 2-step process to ensure patient privacy. The first step was de-identifying all the operative notes to make sure there was no HIPAA-protected information in them.

The second step is one of the key benefits of using open-source models: We can download, edit, and fine-tune them locally without the data ever leaving our institution or being shared with the company that developed the model.

This contrasts with, for example, ChatGPT, where input data is sent back to OpenAI, which would pose serious privacy concerns. So, this 2-step approach allowed us to keep all the data confined within our institution.

As for your second point about whether AI models generate different outputs when given the same input multiple times, many language models have a parameter called "temperature," which controls the level of randomness in the output. When the temperature is set to 0, the model behaves deterministically, meaning it will generate the same response every time it sees the same input. The classification model we built works in a similar way. The output is deterministic, so if you input the same operative note multiple times, it will consistently generate the same CPT code predictions.